What is kube-apiserver?

Professional Answer

The API Server is the central management component that exposes Kubernetes APIs, validates requests, enforces authentication and authorization, and acts as the gateway to the cluster.

Unprofessional Answer

API server talks to Kubernetes.

Why is API Server called the “front door” of Kubernetes?

Professional Answer

All internal and external communication in Kubernetes goes through the API Server, making it the single entry point for cluster interactions.

What is kube-apiserver and what does it do?
kube-apiserver is the central management component of Kubernetes that exposes the REST APIs.
All external and internal communication in the cluster goes through the API server.
It handles authentication, authorization, admission control, and request validation.
The API server stores and retrieves cluster state from etcd.
It is stateless and can be horizontally scaled for high availability.

Kubernetes Internals – Component-by-Component

1. kube-apiserver (The Brain Gateway)

What it is internally

- A **stateless REST API service**
- Written in **Go**
- Fronted by HTTPS
- Can be horizontally scaled

Internal Responsibilities

1. **Authentication**
    - Validates *who* is making the request

- Supports:
  - Certificates
  - Tokens
  - OIDC
  - Service Accounts

2. **Authorization**
   - Decides *what* the user can do
   - Uses:
     - RBAC
     - ABAC
     - Webhooks
3. **Admission Controllers**
   - Intercepts requests **after auth**
   - Can:
     - Validate objects
     - Mutate objects
   - Examples:
     - LimitRanger
     - ResourceQuota
     - PodSecurity
     - MutatingWebhook

4. **API Validation**
    - Validates schema
    - Ensures object correctness
5. **Persistence**
    - Writes state to **etcd**
    - Reads state from etcd

Key Internal Concept

kube-apiserver is the **only component that talks to etcd**

Alright, let's go **full senior-level on kube-apiserver**.
This is the **most important Kubernetes component in interviews**—and in real outages.

🔥 **kube-apiserver – The Front Door of Kubernetes**

1. What is kube-apiserver?

The kube-apiserver is the **central management component** of Kubernetes.
It exposes the **Kubernetes API** over REST/HTTPS.
All cluster operations go through it.
It validates, authenticates, authorizes, and persists data.
No other component talks directly to etcd.

2. Why kube-apiserver is critical

- Single entry point to the cluster
- Controls state transitions
- Enforces security policies
- Mediates between users and control plane
- Failure = control plane outage

👉 **If kube-apiserver is down, Kubernetes is down.**

3. Who talks to kube-apiserver?

Component	Interaction
kubectl	REST calls
Scheduler	Watches Pods
Controller Manager	Watches desired vs actual
kubelet	Pod status updates
etcd	Read/write state

Rule: Nothing bypasses the API Server.

4. Internal request flow (exam-ready)

1. Client sends HTTPS request
2. Authentication (cert, token, OIDC)

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3. Authorization (RBAC, ABAC)
4. Admission Controllers
5. Schema validation
6. Persist to etcd
7. Notify watchers

This order **always comes in interviews**.

5. Authentication in kube-apiserver

Supports:

- Client certificates
- Bearer tokens
- Service accounts
- OIDC (IAM, LDAP, AD)

Authentication answers **WHO are you**.

6. Authorization in kube-apiserver

Common modes:

- RBAC (most used)
- ABAC
- Webhook
- Node authorization

Authorization answers **WHAT can you do**.

7. Admission Controllers (Very Important)

Admission controllers modify or reject requests.

Types:

- **Mutating** (modify request)
- **Validating** (approve/reject)

Examples:

- LimitRanger
- ResourceQuota
- PodSecurity

- ImagePolicyWebhook

👉 Runs **after auth, before etcd write**.

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8. Why kube-apiserver is stateless

- No local state stored
- Can run multiple replicas
- HA via load balancer
- Easy scaling

State is always in etcd.

9. kube-apiserver HA design

- Multiple replicas
- Fronted by Load Balancer
- All connect to same etcd cluster
- No leader election needed

👉 Scaling API Server ≠ scaling etcd.

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10. What happens if kube-apiserver goes down?

- kubectl stops working
- No scheduling or scaling
- Controllers stop reconciliation
- Existing Pods keep running

Looks alive, but frozen.

11. Common kube-apiserver failure causes

- etcd slow/unavailable
- High request rate (CI/CD storms)
- Admission controller misconfig
- Certificate expiry
- Resource starvation

12. Real production outage example

Incident

- Massive GitOps sync loop
- Thousands of API calls/sec

Impact

- API Server CPU 100%
- etcd overloaded
- Cluster unusable

Fix

- API rate limiting
- Reduced polling
- Horizontal scaling

13. How to troubleshoot kube-apiserver

kubectl get componentstatuses

kubectl logs kube-apiserver-<node>

kubectl get --raw /healthz

kubectl get --raw /readyz

Check:

- Latency
- Error rates
- TLS errors

14. kube-apiserver ports

Port	Purpose
6443	Secure API
8080	Insecure (deprecated)

Never expose insecure port.

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15. kube-apiserver vs etcd (common confusion)

kube-apiserver	etcd
API gateway	Data store
Stateless	Stateful

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kube-apiserver	etcd
Enforces security	Stores state
Scales horizontally	Needs quorum

16. Ultra-hard interview question

Q: Why doesn't kubelet talk directly to etcd?

Answer:
To enforce centralized validation, security, and consistency.
Direct etcd access would bypass RBAC and admission controls.
API Server acts as a policy gatekeeper.

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17. Professional vs Unprofessional answer

Unprofessional:

API Server manages Kubernetes.

Professional:

kube-apiserver is the authoritative control-plane component that validates, secures, and persists all cluster state transitions by mediating between clients, controllers, and etcd.

18. Exam one-liner

kube-apiserver is the secure, stateless front end that exposes Kubernetes APIs and coordinates all cluster state changes.

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19. Red flags interviewers look for

- 🚩 "kubelet talks to etcd"
- 🚩 "API Server stores data"
- 🚩 "If API Server restarts, Pods die"

(All are wrong.)

20. Senior-level golden rule

If Kubernetes feels slow, blame kube-apiserver first and etcd second.